2013 IEEE 9th International Colloquium on Signal Processing and its Applications, 8 - 10 Mac. 2013, Kuala Lumpur, Malaysia

Fast and Efficient Document Image Clean Up and Binarization Based on Retinex Theory

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***Abstract*— Conversion from gray scale or color document image into binary image is the main and important step in most of optical character recognition (OCR) systems and document analysis. Most of the document images after digitization often suffer from poor contrast, noise, uniform lighting, and shadow. Clean up and binarization is an active subject in image processing, which addresses these problems. Most of the previous binarization methods that depend on local thresholding consume more time. The other methods that depend on global threshold are fast but don't work well when the document image is degraded. In this paper we present fast and efficient document image clean up and binarization method based on retinex theory and global threshold. The proposed method is fast and produces high quality results compared to the previous works.**

Index Term – Binarization, Thresholding, Retinex theory.

I. INTRODUCTION

Document image clean up and binarization is an interesting research topic in image processing that used in most of document image analysis and retrieval. The degree of in character segmentation and recognition is depends on the binarization of document image. In recent years, transforming cultural and historical documents into digital format has become an important trend. Most of old document images that have historical value after digitization are very poor and often contain unstable luminance. Old document images are usually subject to degradation due to uniform lighting from camera, dust and dirt on the glass of the scanner, poor storage, and bad environment [1]. Fig. 1 shows an example of degraded and poor quality document image.

The purpose of document image clean up and binarization is to convert the poor quality gray or color document image which suffering from degradation to binary form and makes it very clear to be readable by human or OCR systems. Most of OCR, document analysis, and document understanding systems have been designed to work on bi-level document image. So, binarization is the first step in most of these systems. Many thresholding techniques have been developed in the past which deal with degraded and poor quality document image but it is still difficult to deal with document images with very low quality due to variable background intensity, very low local contrast, smear, smudge and shadows [2].

Fig.1 Example of degraded and poor quality document images

In general all techniques which deal with document image binarization are classified into local and global. In global techniques, single threshold value is selected for the whole document image as in [3-7]. Actually these techniques are efficient to convert any grayscale image into a bi-level one if the document image has good separation between background and foreground. They are inappropriate for degraded documents.

Global binarization methods tend to produce marginal noise along the page borders. Also in the smoothed grey-scale histogram, the structure of the peaks, valleys and curvatures are used to determine the final global threshold value as in [8, 9].

The common disadvantage of histogram-based global methods is their ineffectiveness for document images that do not have a bimodal histogram distribution, even when the images contain clear handwriting on a contrasting background. To overcome these problems, local thresholding methods have been proposed for document binarization. They compute the threshold individually for each pixel by using information from the local neighborhood of the pixel.

In [10, 11] the authors presented a local binarization method that calculates the local threshold using the mean and standard deviation of pixels within a local neighborhood window. The performance of these methods is heavily depends on the size of the local neighborhood window. T.Romen et al [12] pioneered a fast local adaptive thresholding method that extends the method in [10, 11] to minimize the computational time by using local mean and mean deviation.

J. G. Kuk et al [13] define descriptor that measures the variability of pixel values around a given pixel to classify the pixel into text, near text and background pixel. Graph cut energy minimization are used to relabeled the noise pixels. Kim et al [14] proposed adaptive local thresholding technique based on 3D terrain and water flow model. The valleys that corresponding to the region is locally detected by the water flow model. Finally, the global threshold method in [8] is used to differentiate between the original and the water-filled terrain.

Shijian et al [15] proposed a local thresholding method based on document background surface and the text stroke edge. Document background surface is estimated using an iterative polynomial smoothing method. The stroke edges are then detected based on the local image variation within the compensated document image. Finally, the local threshold is estimated based on the detected stroke edge pixels within a local neighborhood window.

In general, local adaptive methods achieve better results than global ones but most of the local thresholding document image binarization techniques suffer from the following major limitations:

- Region size dependant.
- Individual image characteristics
- Time consuming.

In this paper, we propose fast and efficient clean up and global thresholding method which makes enhancement on degraded and poor quality document image before binarizes the image. The proposed algorithm based on retinex theory which treats the degradation problems of the document image, and global thresholding binarization to convert the document image into binary form.

The rest of this paper is arranged as follows. Section 2 presents the proposed scheme in details. Section 3 illustrates some examples to show the effectiveness of the proposed method. Finally the conclusion is drawn in section 4.

II. PROPOSED METHOD

Most of the previous binarization methods that depend on local thresholding produce high quality results but consume more time. The other methods that depend on global threshold are fast but don't work well when the document image is degraded. Our proposed binarization method combines the advantages of local and global thresholding by using the concept of retinex theory which can effectively enhance the degraded and poor quality document image. After that, fast global threshold is used to convert the document image into binary form.

The proposed document clean up and binarization method is based on two main steps to convert the document image to binary form. The first is degradation enhancement and the second is a global thresholding. By using these steps, the quality of the result image is improved and the processing time is fast compared to the others methods.

A. Degradation Enhancement

The concept of retinex theory comes from the biological phenomenon of human visual system. The formation of the retinex theory as in equation 1:

$$I(x, y) = R(x, y) . L(x, y) \quad (1)$$

Where *I (x,y) is* the intensity of the image with reflectance *R (x, y)* and illumination *L (x, y)* which can approximated by using low-frequency component of the measured image [17]. So, the proposed method uses the retinex theory. If we have grayscales image I(x, y) has degradation, we can obtain the Lightness image L(x,y) by divide the original grayscale image I(x, y) on its smooth version M(x, y) as the following equation :

$$L(x, y) = {I(x, y)}/{M(x, y)} \quad (2)$$

Most common binarization methods like [13] use Gaussian filter with large kernel as a smoothed version but the result is not clear and still have some limitations. The proposed work uses the median filter instead of Gaussian to obtain most smooth version. The median filter operates over a window by selecting the median intensity in the window. Median filter is more robustness to outliers, noise removal, and gives more smooth result compared to Gaussian filter.

Fig. 2 shows a comparison between Gaussian and median filter in lightness document image. Fig. 3 illustrates the effectiveness of the first step of the proposed method on the degraded image in fig. 1.

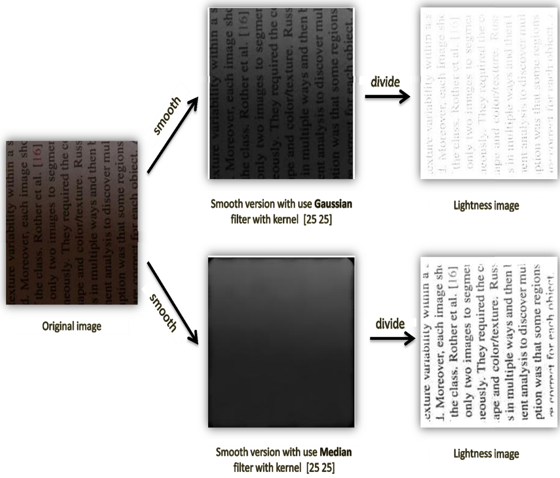

Fig. 2 Comparison between the proposed median filter and Gaussian filter in [13].

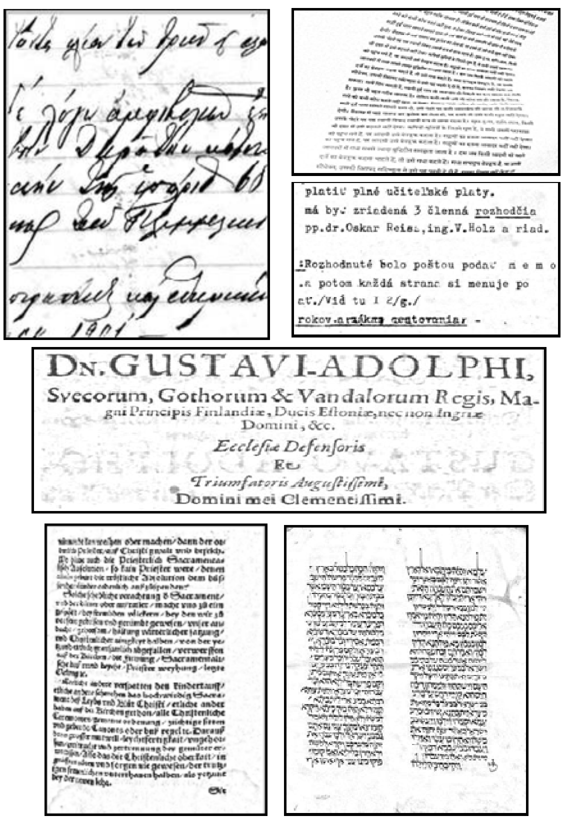

Figure 3 Degraded document images in figure 1 after Enhancement step.

B. Global Thresholding

Most of the methods that depend on global threshold are very fast but don't work well and tend to border noise when the document image is degraded or has illumination. The first step of the proposed work overcomes this limitation by removing the illumination and the degradation before performing global thresholding. The global thresholding becomes easier and accurate after degradation enhancement step. The most common global thresholding method in [8] is used to convert the lightness image into bi-level one (black for text and white for background). Fig. 4 illustrates comparison of the binarization using the proposed method and the method in [8]. It is obvious that large dark area appear in the results of method [8] due to the bad illumination of the original image. While our method, the results are more clear and readable for human and OCR systems.

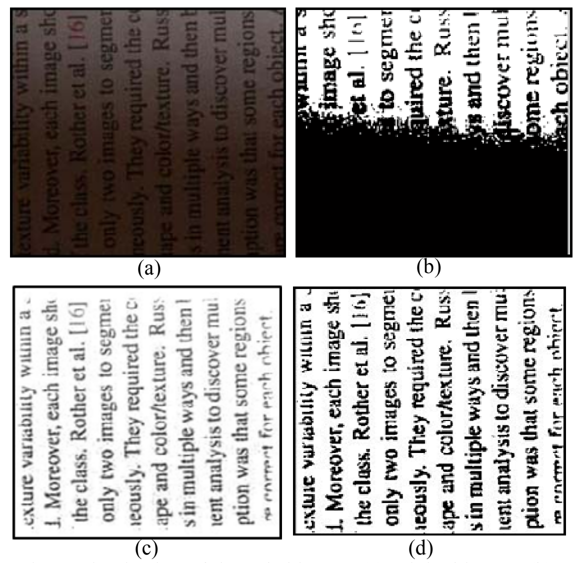

Fig. 4 Binarization of degraded image. (a) original image, (b) binarization using method [8], (c) degradation enhancement, (d) binarization using proposed method.

III. DISCUSSION OF RESULTS

The proposed method has been implemented using MATLAB R2009a and tested on a PC with Pentium Dual Core 1.8 GHz CPU. To demonstrate its effectiveness, it is tested on a variety of degraded and noisy document images. Each document image illustrates the efficiency of the proposed method in addressing one of the binarization challenging like low contrast, bad illumination, double side noise, and smeared documents.

Fig. 5 presents the results of the proposed binarization method on a collection of degraded document images shown in fig. 1. Some of these images are smeared handwritten document, bad illumination and others have double side noise. Based on the visual criteria, the proposed method works well and produce high quality and readable results in all cases of degradation.

For more experimental results the proposed method is compared with the most three state of art methods Otsu's method [8], Niblack's method [10], Sauvola method [11], and other recently methods [14,18,19]. Fig. 6 and 7 illustrate the results of the proposed work and the others methods on poor quality historical handwritten document image. Based on the visual criteria our proposed method is efficient and more readable than others methods.

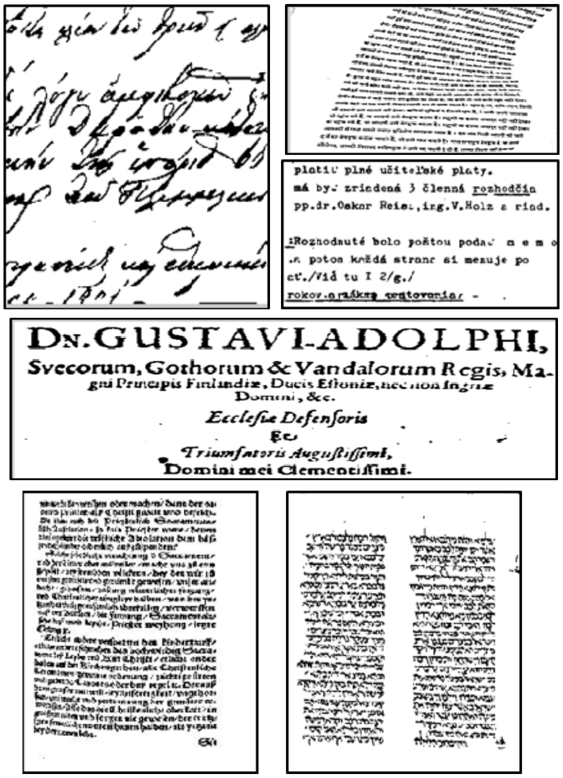

Fig. 5 Proposed binarization results of the document image in fig. 1.

(a) (b) (c)

(d) (e) (f)

Fig. 6 Binarization of poor quality historical handwritten document image. (a) original image, (b) Otsu's method [8], (c) Niblack's method [10], (d) Sauvola method [11], (e) Kim method [14], and (f) our proposed method.

(a) (b) (c)

(d) (e) (f)

Fig. 7 Binarization of historical document. (a) Original image, (b) Otsu's method [8], (c) Niblack's method [10], (d) Y. Yang method [19], (e) Bernsen's method [18], (f) our proposed method.

IV. CONCLUSIONS

In this work, we proposed a fast and efficient document image clean up and binarization method based on retinex theory and global thresholding. The proposed method combines the advantages of local and global thresholding by using the concept of retinex theory which can effectively enhance the degraded and poor quality document image. After that, fast global threshold is used to convert the document image into binary form. The proposed method overcomes the limitations of the related global threshold techniques. In addition to, it work well when the document image have any type of degradation.

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